**Experiment- 10:**

**OBJECTIVE:**

To distinguish the responses produced by initial stored energy and external excitation, obtain the impulse response of a continuous-time LTI system, and verify that the total response is the sum of the zero-input and zero-state responses, with the zero-state response also obtained through convolution.

**Experimental Procedure:**

1. Define the continuous-time system dy(t)/dt + 2y(t) = x(t) and obtain H(s) under zero initial conditions.

2. Use MATLAB to display the transfer function and its pole.

3. Plot the impulse response and identify its decay pattern.

4. Obtain the zero-input response for y(0−) = 1 with x(t) = 0.

5. Obtain the zero-state response to x(t) = u(t) with zero initial conditions.

6. Compute the convolution of the unit-step input and impulse response, then compare it with the zero-state response.

7. Combine the zero-input and zero-state responses and verify the total response.

**Code:**

```
clc;

clear;

close all;


% Experiment 10: System Responses

% System: dy(t)/dt + 2y(t) = x(t)


num = [1];

den = [1 2];

sys = tf(num, den);


disp('Transfer Function:');
```

```
sys

disp('Pole:');
disp(pole(sys));

t = 0:0.01:5;
dt = t(2) - t(1);

% Impulse Response
h = exp(-2*t);

figure;
impulse(sys, t);
grid on;
title('Impulse Response h(t)');
xlabel('Time (s)');
ylabel('h(t)');

% Zero-Input Response (Initial condition = 1)
y_zi = exp(-2*t);

figure;
plot(t, y_zi, 'LineWidth', 1.5);
grid on;
title('Zero-Input Response');
xlabel('Time (s)');
ylabel('y_{zi}(t)');
```

```
% Zero-State Response to Unit Step
x_step = ones(size(t));
[y_zs, tout] = step(sys, t);
y_zs = squeeze(y_zs);

figure;
plot(tout, y_zs, 'LineWidth', 1.5);
grid on;
title('Zero-State Response');
xlabel('Time (s)');
ylabel('y_{zs}(t)');

% Convolution
y_conv = conv(x_step, h) * dt;
y_conv = y_conv(1:length(t));

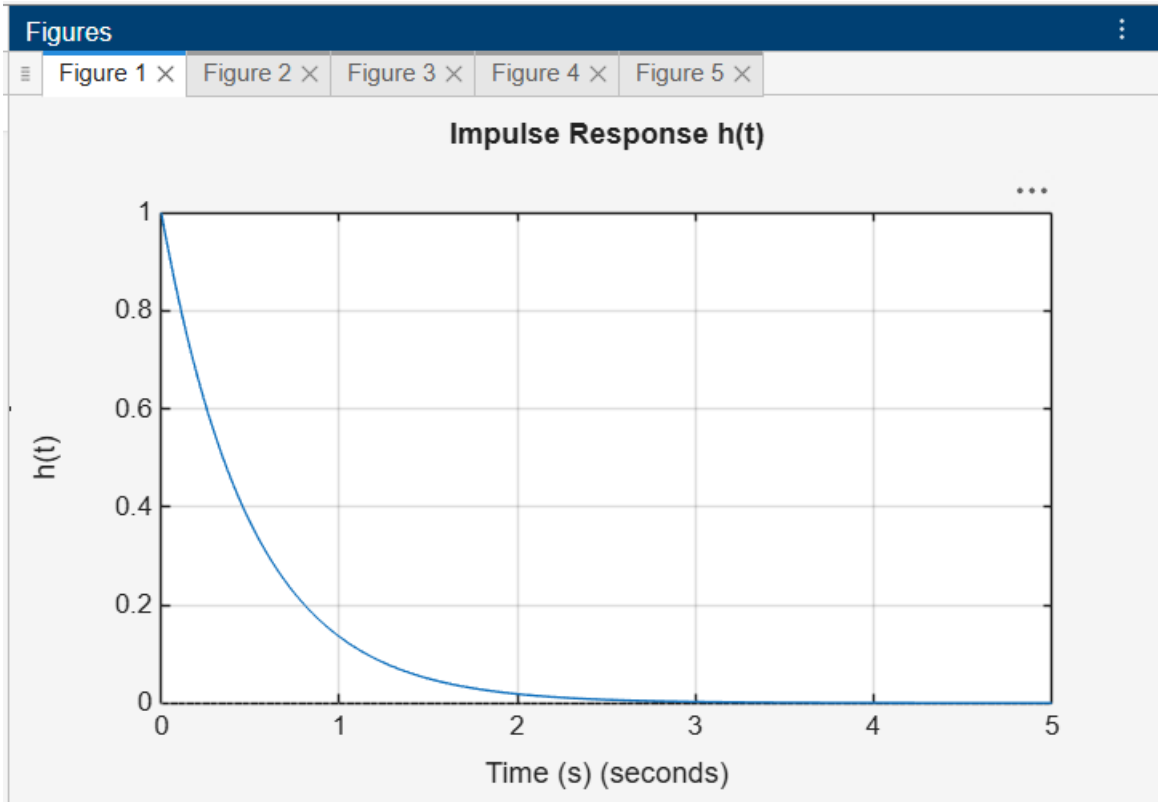
% Total Response
y_total = y_zi + y_zs(:).';

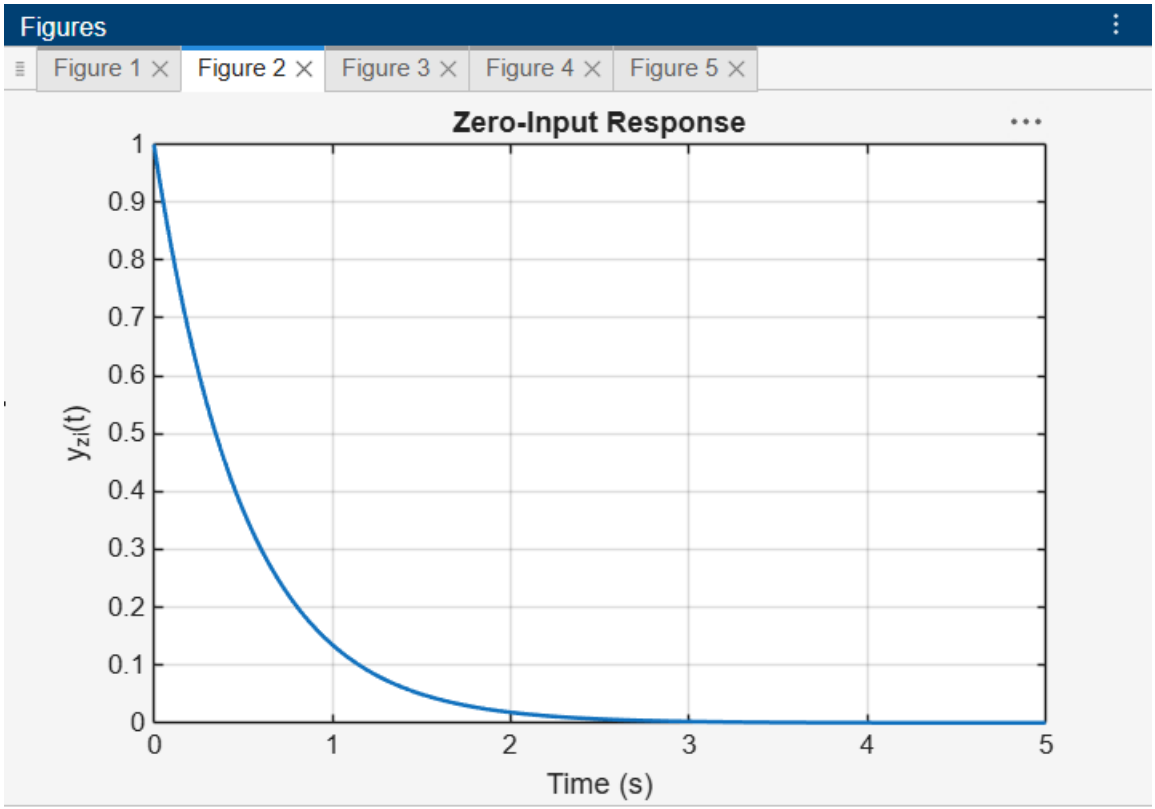
figure;
plot(t, y_zi, 'r', t, y_zs(:).', 'b', ...
   t, y_total, 'k', 'LineWidth', 1.5);
grid on;
legend('Zero-Input', 'Zero-State', 'Total Response');
title('Comparison of Responses');
xlabel('Time (s)');
ylabel('Response');
```

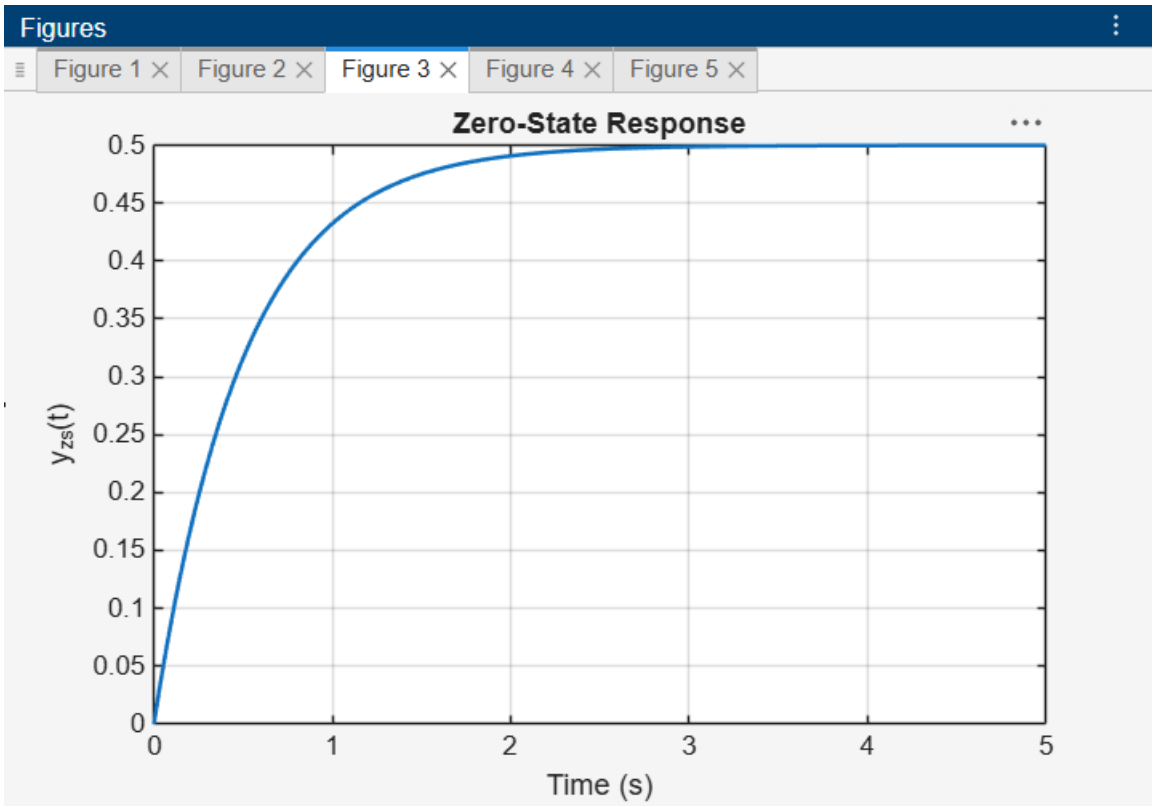
```
% Convolution Verification
y_lsim = lsim(sys, x_step, t);

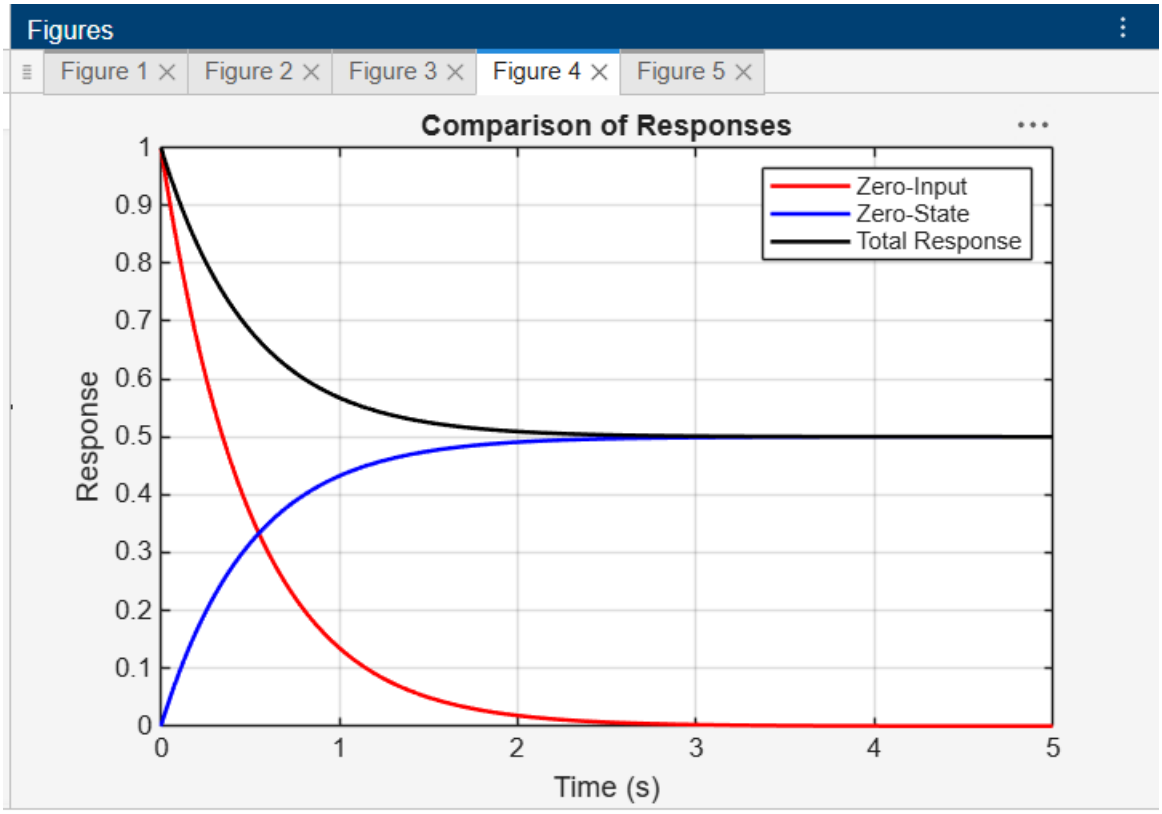

figure;
plot(t, y_conv, '--r', t, y_lsim(:).', 'b', ...
   'LineWidth', 1.5);
grid on;
legend('Convolution', 'LSIM Response');
title('Convolution Verification');
xlabel('Time (s)');
ylabel('y_{zs}(t)');
```

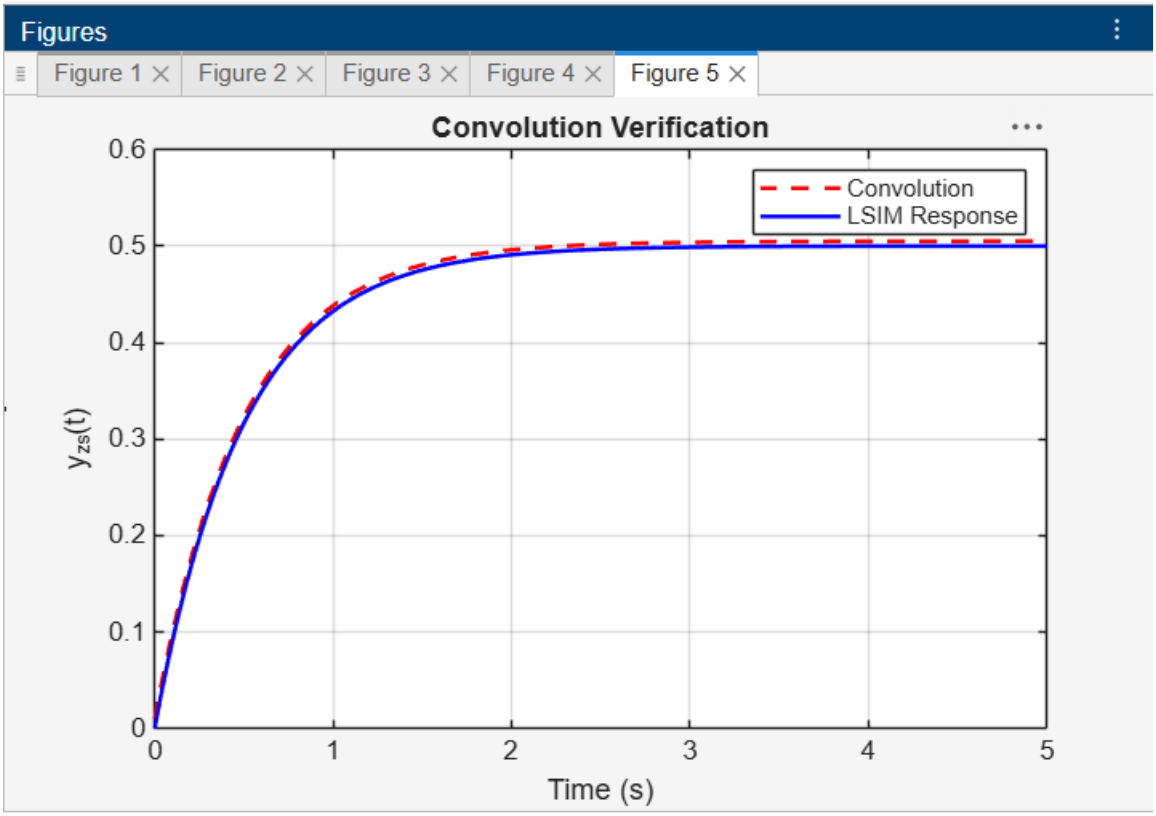
**output:**




Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Zero-Input Response
$y_{zi}(t)$
Time (s)


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Zero-State Response
$y_{zs}(t)$
Time (s)


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Comparison of Responses
Zero-Input
Zero-State
Total Response
Response
Time (s)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
1
2
3
4
5


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Convolution Verification
Convolution
LSIM Response
$y_{zs}(t)$
Time (s)
0
0.1
0.2
0.3
0.4
0.5
0.6
0
1
2
3
4
5

**Result:**

Thus, the zero-input, zero-state, and impulse responses of the continuous-time LTI system were successfully obtained using MATLAB. The total response was verified as the sum of the zero-input and zero-state responses. The zero-state response was also obtained through convolution, confirming the system's linearity and superposition property.

**LABORATORY PERFORMANCE ASSESSMENT RUBRIC**

| Criterion | Excellent – 4 | Good – 3 | Satisfactory – 2 | Needs Improvement – 1 | Weightage |
|---|---|---|---|---|---|
| System modelling and transfer function | Correct model and H(s) | Minor notation error | Partial model | Incorrect model | 20 |
| Response analysis | Correctly distinguishes all responses | Mostly correct | Partial distinction | Major confusion | 20 |
| MATLAB code and execution | Correct, clear, reproducible | Minor code issue | Runs with help | Does not run | 20 |
| Graphs and interpretation | All plots labelled and interpreted | Plots mostly correct | Incomplete plots | Missing/incorrect plots | 15 |
| Convolution verification | Correct convolution and comparison | Small numerical discrepancy explained | Partial verification | No valid verification | 15 |
| Record, observations and conclusion | Complete and technically sound | Minor omissions | Several omissions | Incomplete | 10 |

| | | | | | |
|---|---|---|---|---|---|
| Total | | | | | 100 marks |